Optimization TA 5

Kacper Krasowski

Exercise 5.5

Problem

Let $\phi : \mathbb{R} \rightrightarrows [0,1]$ be defined by $\phi(x) = \{0,1\}$ if $x \leq 0$ and $\phi(x) = \{1\}$ otherwise. Check whether it is uhc, lhc.

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Is ϕ upper hemicontinuous? Lower hemicontinuous?

Exercise 5.5: Definitions

ϕ upper hemicontinuous? Lower hemicontinuous?

Definitions

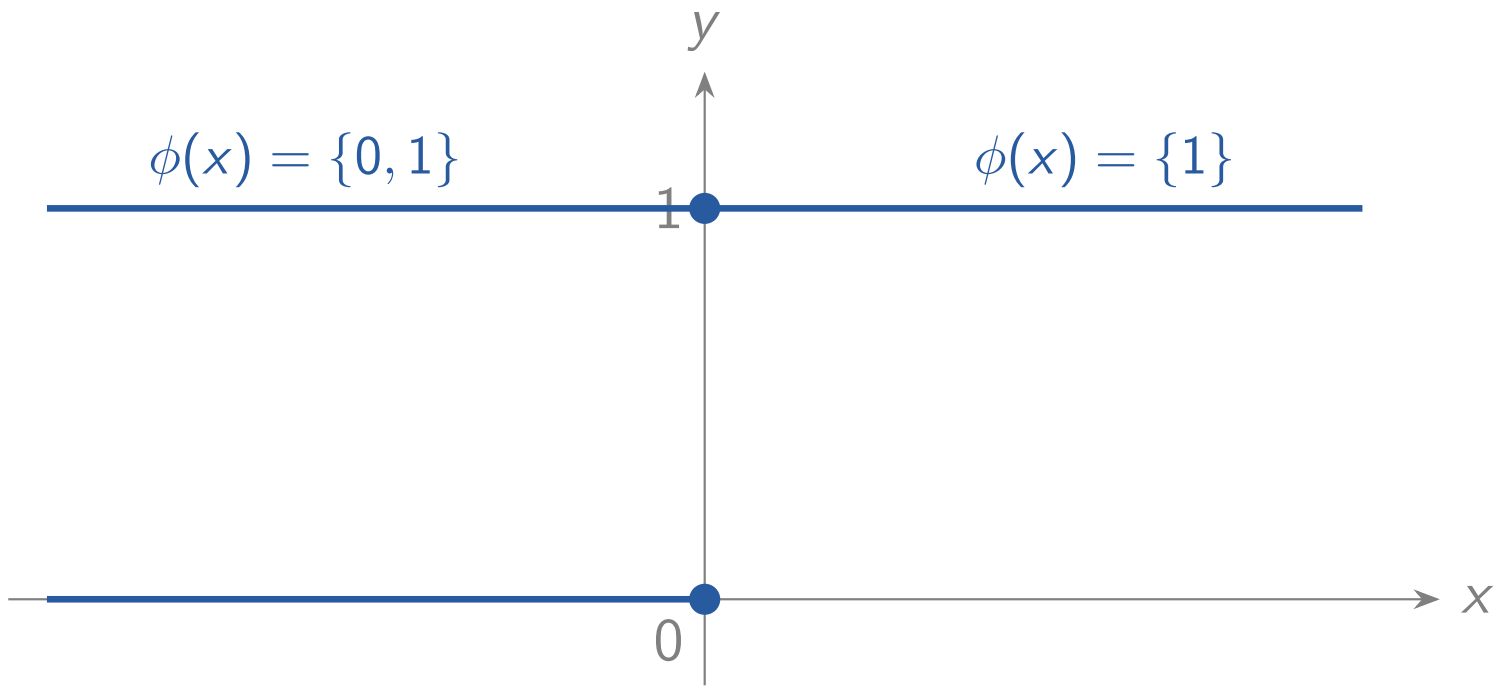
Upper hemicontinuous (uhc) at x: A correspondence $\phi : X \rightrightarrows Y$ is uhc at $x \in X$ if for every open set $S \subseteq Y$ containing $\phi(x)$, there exists $\delta > 0$ such that

$$\phi(B_\delta(x)) \subseteq S.$$

Lower hemicontinuous (lhc) at x: A correspondence $\phi : X \rightrightarrows Y$ is lhc at $x \in X$ if for every open set $T \subseteq Y$ with $\phi(x) \cap T \neq \varnothing$, there exists $\delta > 0$ such that

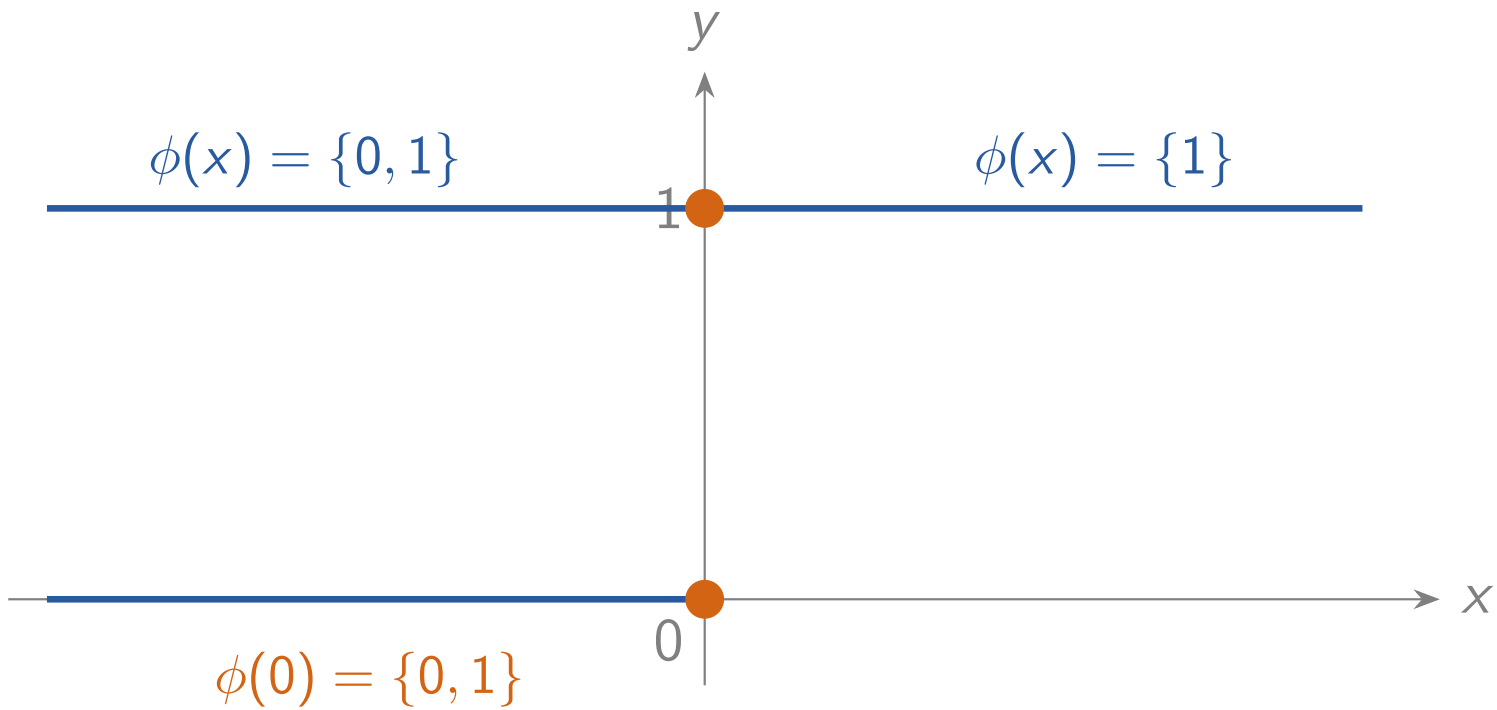
$$\phi(x') \cap T \neq \varnothing \quad \text{for every } x' \in B_\delta(x).$$

Exercise 5.5: Intuition



The values are sets: two points on the left, one on the right.

Exercise 5.5: Intuition



At zero, both 0 and 1 belong to $\phi(0)$.

Exercise 5.5: Sketch of the Proof — Uhc

Check upper hemicontinuity on $\mathbb{R}$.

1. Check uhc for $x \leq 0$.

Exercise 5.5: Sketch of the Proof — Uhc

Check upper hemicontinuity on $\mathbb{R}$.

1. Check uhc for $x \leq 0$.
2. Check uhc for $x > 0$.

Exercise 5.5: Sketch of the Proof — Lhc

Disprove lower hemicontinuity at $x = 0$.

1. Negate the definition of lhc at $x_0 = 0$.

▸ Appendix: Open in the codomain

Exercise 5.5: Sketch of the Proof — Lhc

Disprove lower hemicontinuity at $x = 0$.

1. Negate the definition of lhc at $x_0 = 0$.
2. Find an open set for which the condition fails.

▸ Appendix: Open in the codomain

Exercise 5.6

Problem

Show that the budget correspondence $B : \mathbb{R}^3_{++} \rightrightarrows \mathbb{R}^2_{++}$ with

$$B(p_1, p_2, w) = \{(x_1, x_2) \in \mathbb{R}^2_+ : p_1 x_1 + p_2 x_2 \leq w\}$$

is uhc.

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is uhc.

$$\boxed{p_1, p_2, w > 0 \implies B \text{ is uhc at } (p_1, p_2, w)}$$

Exercise 5.6: Definitions and Tools

B is upper hemicontinuous.

Upper hemicontinuity: sequential characterization

A correspondence $\phi : X \rightrightarrows Y$ is uhc if for every sequence $x^n \to x$ and every choice $y^n \in \phi(x^n)$, there exists a subsequence (y^{n_k}) such that

$$y^{n_k} \to y \in \phi(x).$$

If ϕ is compact-valued, this condition is also necessary.

Bounded sequence and Bolzano–Weierstrass

A sequence (x^n) in $\mathbb{R}^2$ is bounded if there exists $M > 0$ such that $|x_i^n| \leq M$ for $i = 1, 2$ and every n.

Every bounded sequence in $\mathbb{R}^2$ has a convergent subsequence.

▸ Appendix: Sequential characterization

Exercise 5.6: Sketch of the Proof

B is upper hemicontinuous.

1. Take arbitrary $q^n \to q$ in $\mathbb{R}^3_{++}$ and $x^n \in B(q^n)$.

Exercise 5.6: Sketch of the Proof

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3. Apply Bolzano–Weierstrass.

Exercise 5.6: Sketch of the Proof

B is upper hemicontinuous.

1. Take arbitrary $q^n \to q$ in $\mathbb{R}^3_{++}$ and $x^n \in B(q^n)$.
2. Show that (x^n) is bounded.
3. Apply Bolzano–Weierstrass.
4. Show that the limit belongs to $B(q)$ and conclude.

Exercise 6.1

Problem

Let's go beyond Berge's Maximum Theorem. Consider a maximization problem with objective function $f : X \times \Theta \to \mathbb{R}$ where f is increasing in θ (not necessarily continuous). The constraint set $G \subseteq X$ is nonempty and does not depend on θ. Show that $v(\theta) = \sup_{x \in G \subseteq X} f(x, \theta)$ is increasing in θ.

Hint: No need to complicate, solved in a few lines by simple inequalities.

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Hint: No need to complicate, solved in a few lines by simple inequalities.

$$\boxed{\theta_1 < \theta_2 \implies v(\theta_1) \leq v(\theta_2)}$$

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1. Take any $\theta_1 < \theta_2$.
2. Compare $f(x, \theta_1)$ and $f(x, \theta_2)$ for the same $x \in G$.

Exercise 6.1: Sketch of the Proof

$$\boxed{\theta_1 < \theta_2 \implies v(\theta_1) \leq v(\theta_2)}$$

1. Take any $\theta_1 < \theta_2$.
2. Compare $f(x, \theta_1)$ and $f(x, \theta_2)$ for the same $x \in G$.
3. Use the definition of supremum to compare the value functions.

Exercise 6.2

Problem

Let $f : X \to X$ be a function. We say $x \in X$ is a fixed point of f if $f(x) = x$. Prove that every continuous $f : [0, 1] \to [0, 1]$ has a fixed point by showing that the function g with $g(x) = f(x) - x$ satisfies $g(x) = 0$ for some $x \in [0, 1]$.

Hint: Think about something you proved in PS5.

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Hint: Think about something you proved in PS5.

Find $c \in [0, 1]$ such that $g(c) = 0$, hence $f(c) = c$.

Exercise 6.2: Definitions and Tools

$$\boxed{\text{Find } c \in [0,1] \text{ such that } g(c) = 0.}$$

Definitions

Fixed point: A point $c \in X$ is a fixed point of $f : X \to X$ if $f(c) = c$.

Continuous on $[0,1]$: A function $h : [0,1] \to \mathbb{R}$ is continuous if, for every $x_0 \in [0,1]$ and every $\epsilon > 0$, there exists $\delta > 0$ such that, for all $x \in [0,1]$,

$$|x - x_0| < \delta \implies |h(x) - h(x_0)| < \epsilon.$$

Intermediate Value Theorem (TA4, Exercise 5.1)

If $h : [a,b] \to \mathbb{R}$ is continuous, it takes every value strictly between $h(a)$ and $h(b)$ at some $c \in (a,b)$.

Exercise 6.2: Sketch of the Proof

$$\boxed{\text{Find } c \in [0,1] \text{ such that } g(c) = 0.}$$

The function $g(x) = f(x) - x$ is continuous.

1. Check the signs of g at the endpoints.

▸ Appendix: Why is g continuous?

Exercise 6.2: Sketch of the Proof

$$\boxed{\text{Find } c \in [0,1] \text{ such that } g(c) = 0.}$$

The function $g(x) = f(x) - x$ is continuous.

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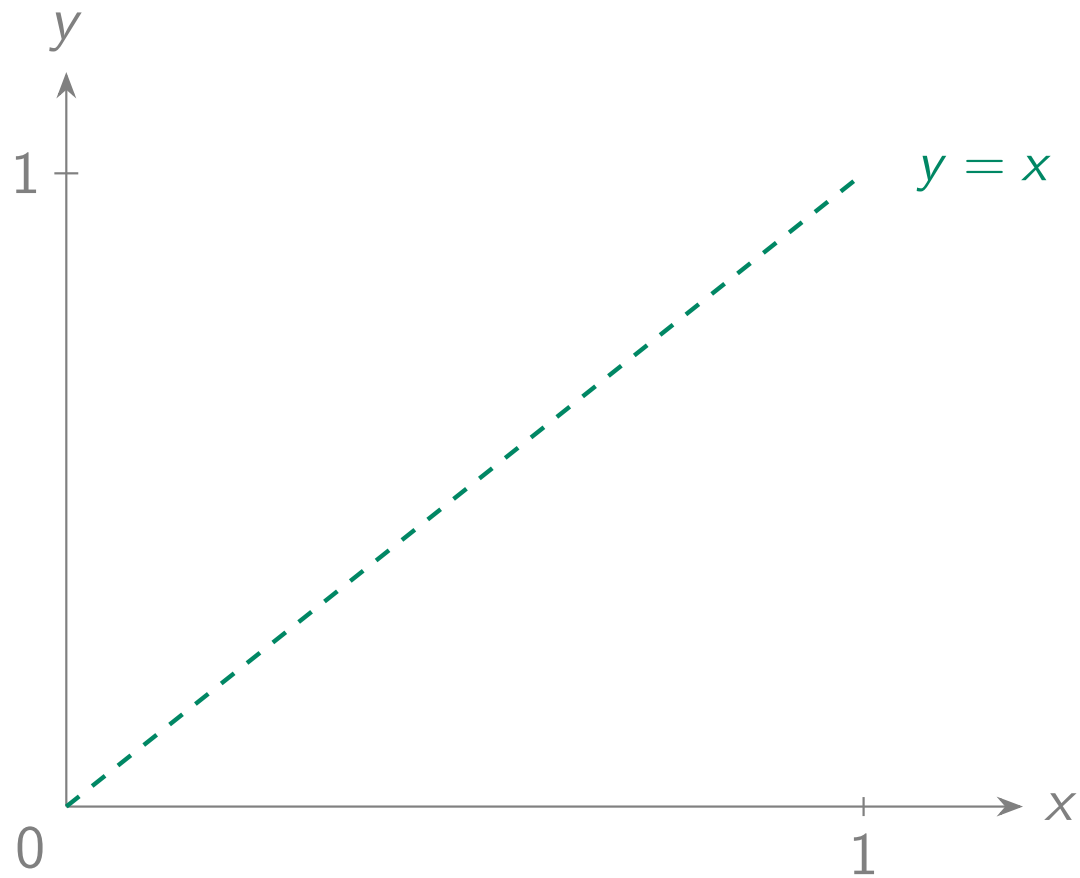
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The function $g(x) = f(x) - x$ is continuous.

1. Check the signs of g at the endpoints.
2. Consider the endpoint cases, then apply the Intermediate Value Theorem.
3. Translate the result into a fixed point of f.

▸ Appendix: Why is g continuous?

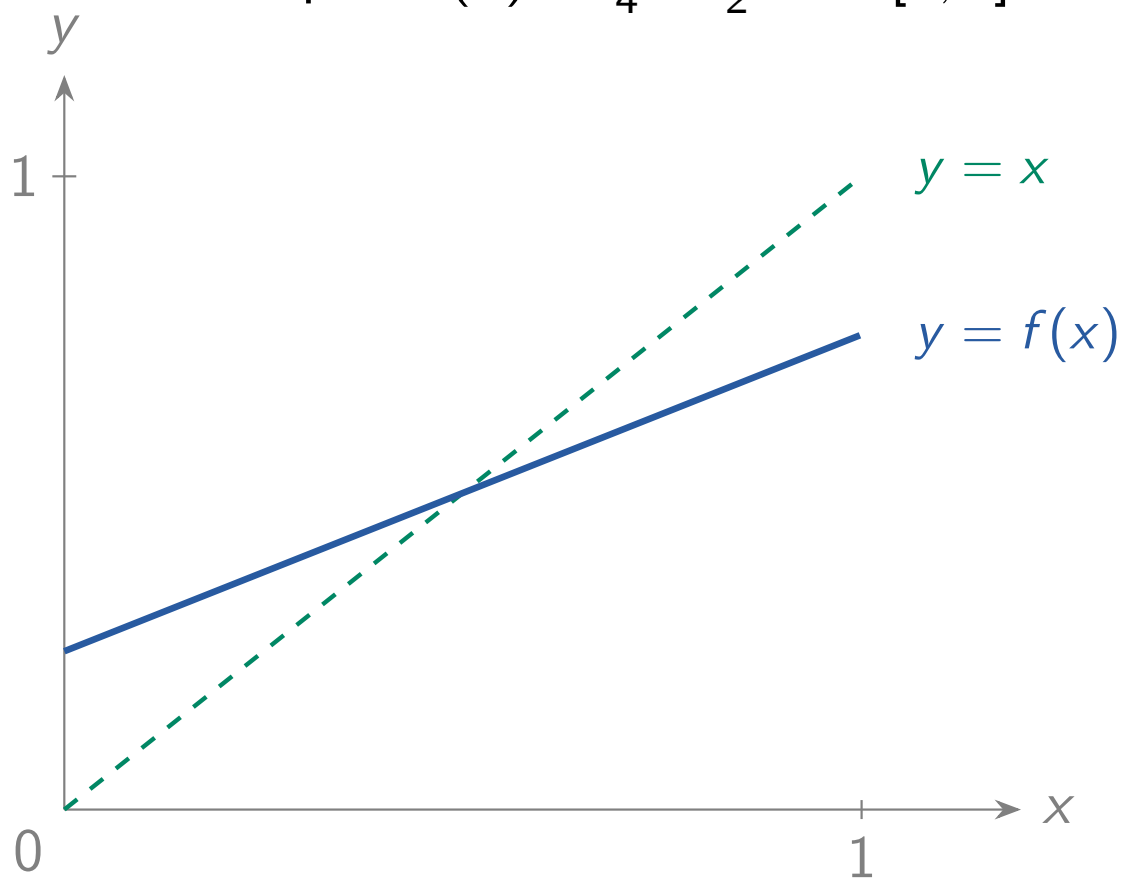
Exercise 6.2: Intuition — Increasing



On the diagonal, the output equals the input.

Exercise 6.2: Intuition — Increasing

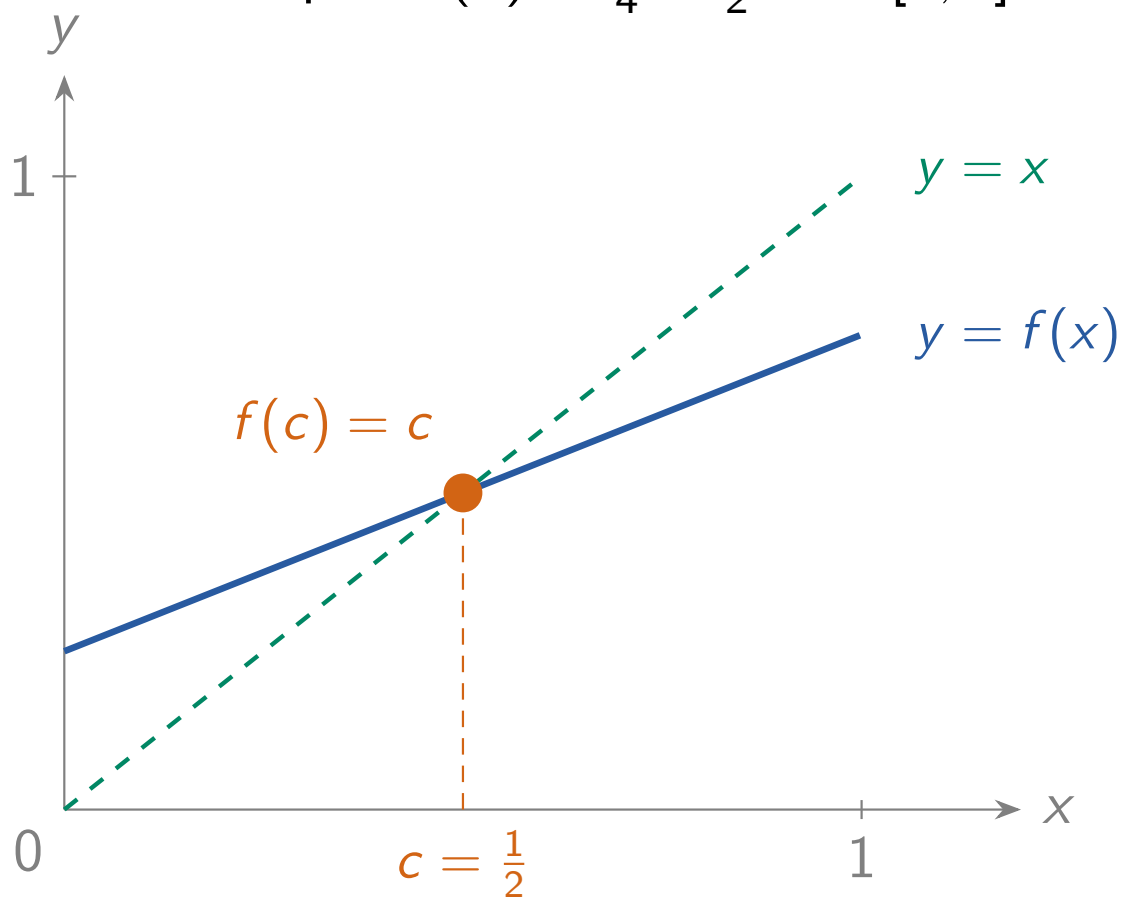
Example: $f(x) = \frac{1}{4} + \frac{1}{2}x$ on $[0, 1]$.



Where does the graph meet the diagonal?

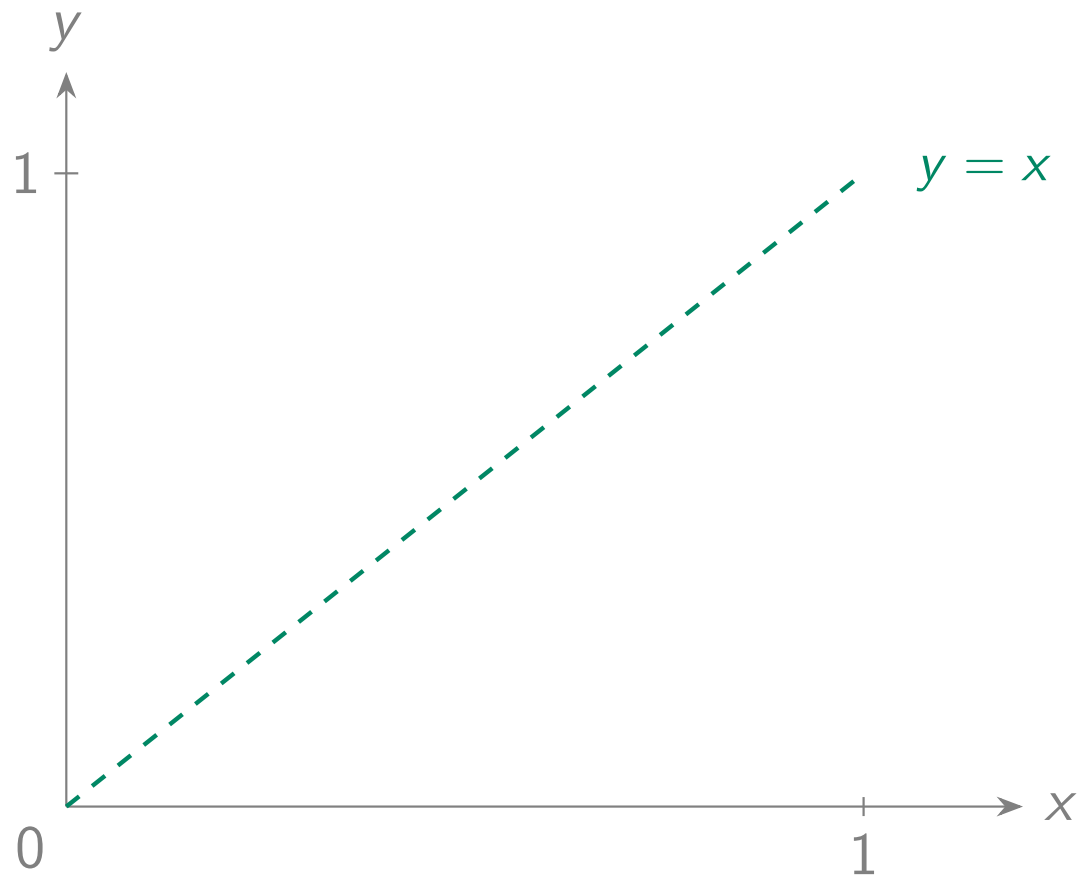
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Example: $f(x) = \frac{1}{4} + \frac{1}{2}x$ on $[0, 1]$.



At the intersection, $f(c) = c$.

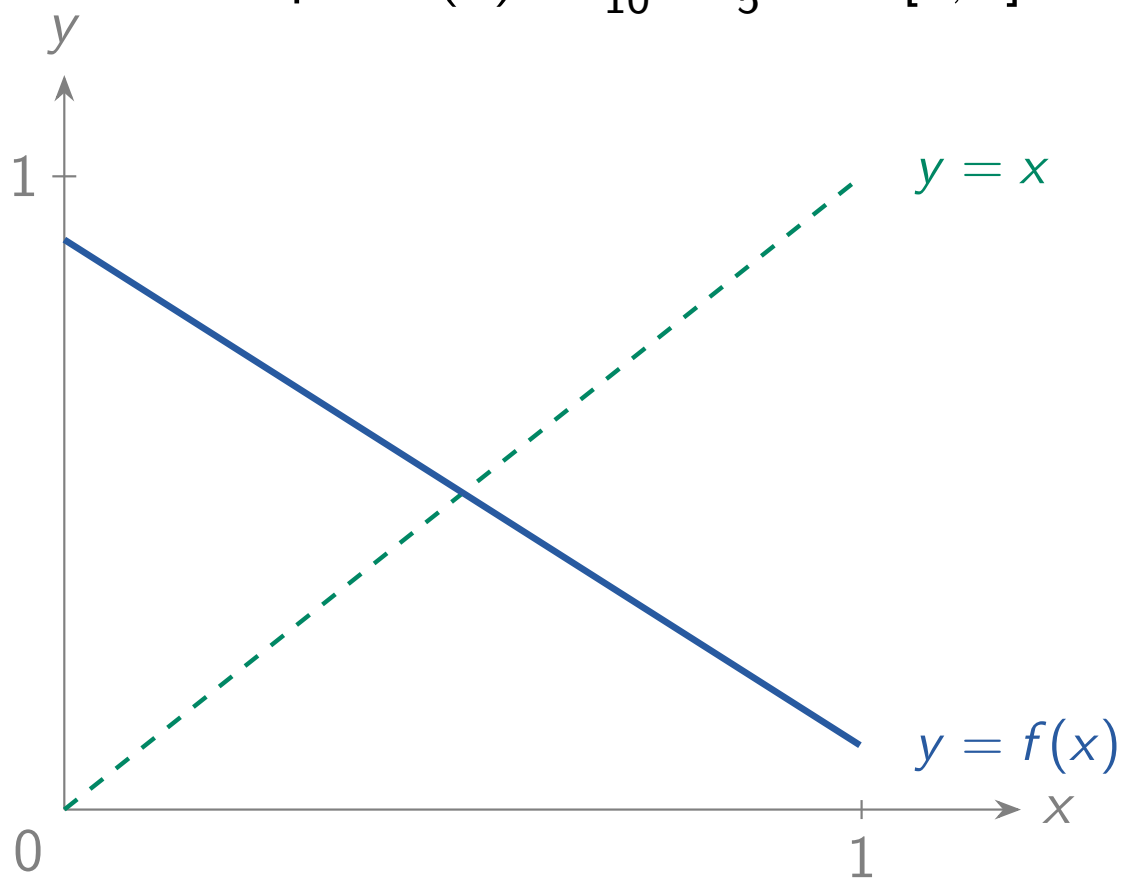
Exercise 6.2: Intuition — Decreasing



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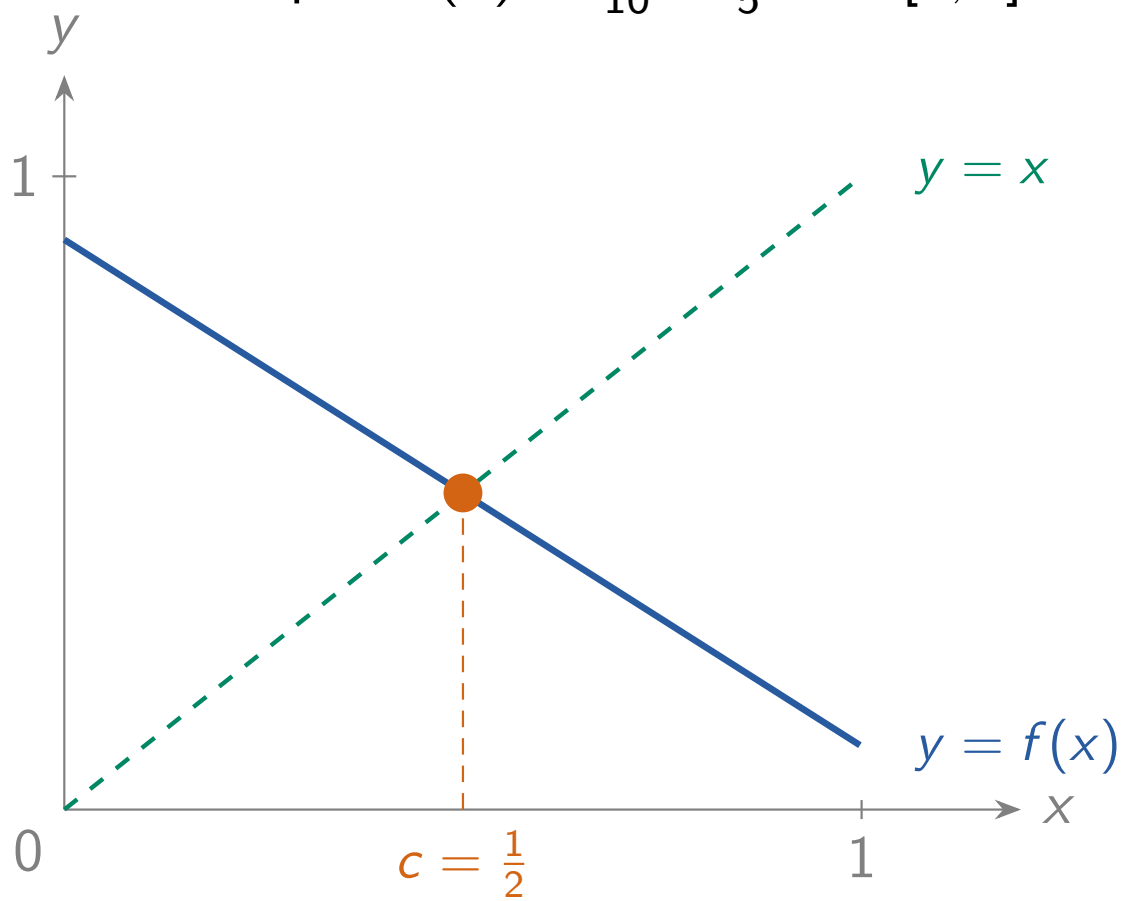
Example: $f(x) = \frac{9}{10} - \frac{4}{5}x$ on $[0, 1]$.



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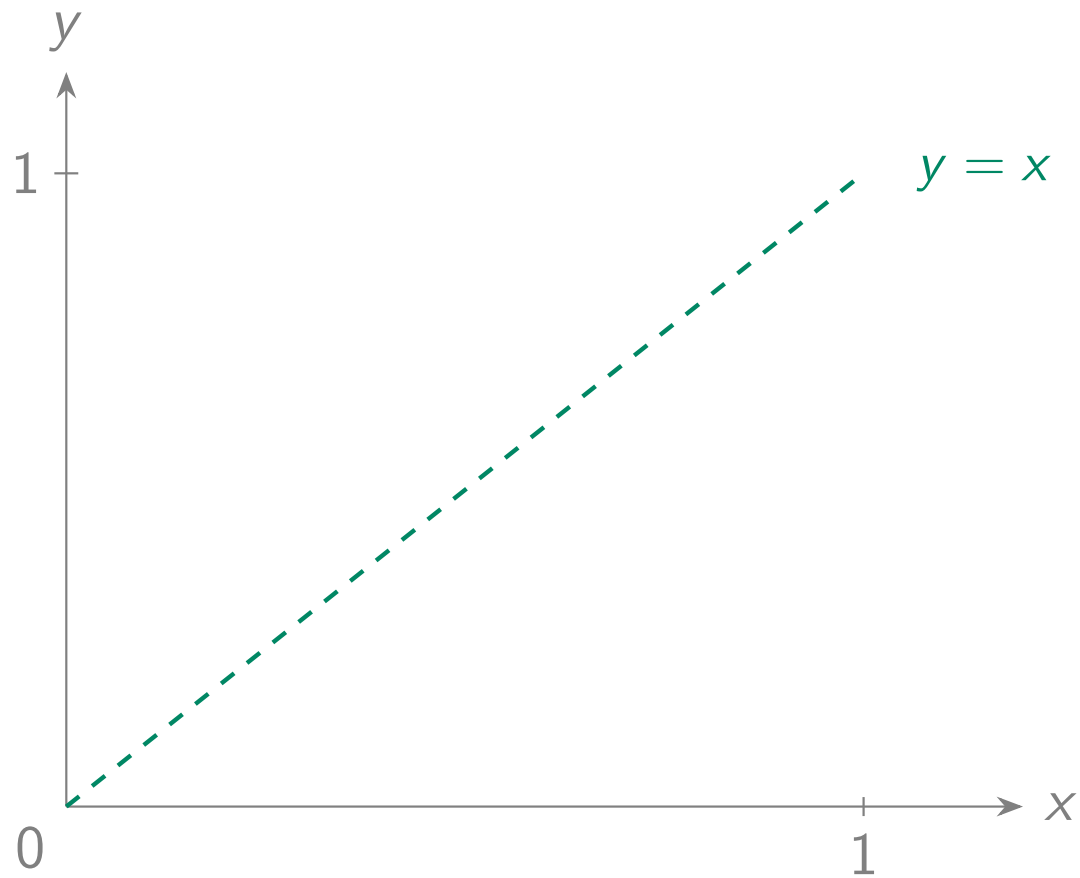
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Example: $f(x) = \frac{9}{10} - \frac{4}{5}x$ on $[0, 1]$.



A decreasing function can also have a fixed point.

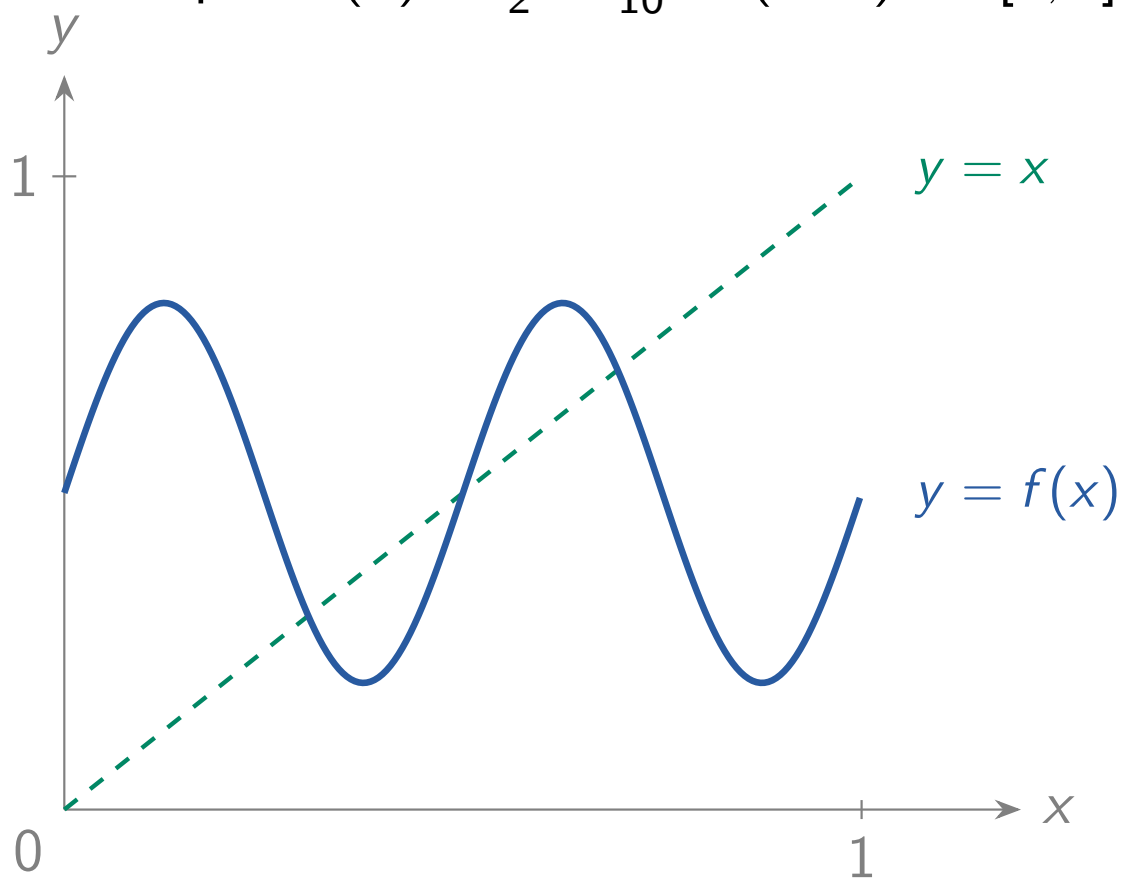
Exercise 6.2: Intuition — Wobbly



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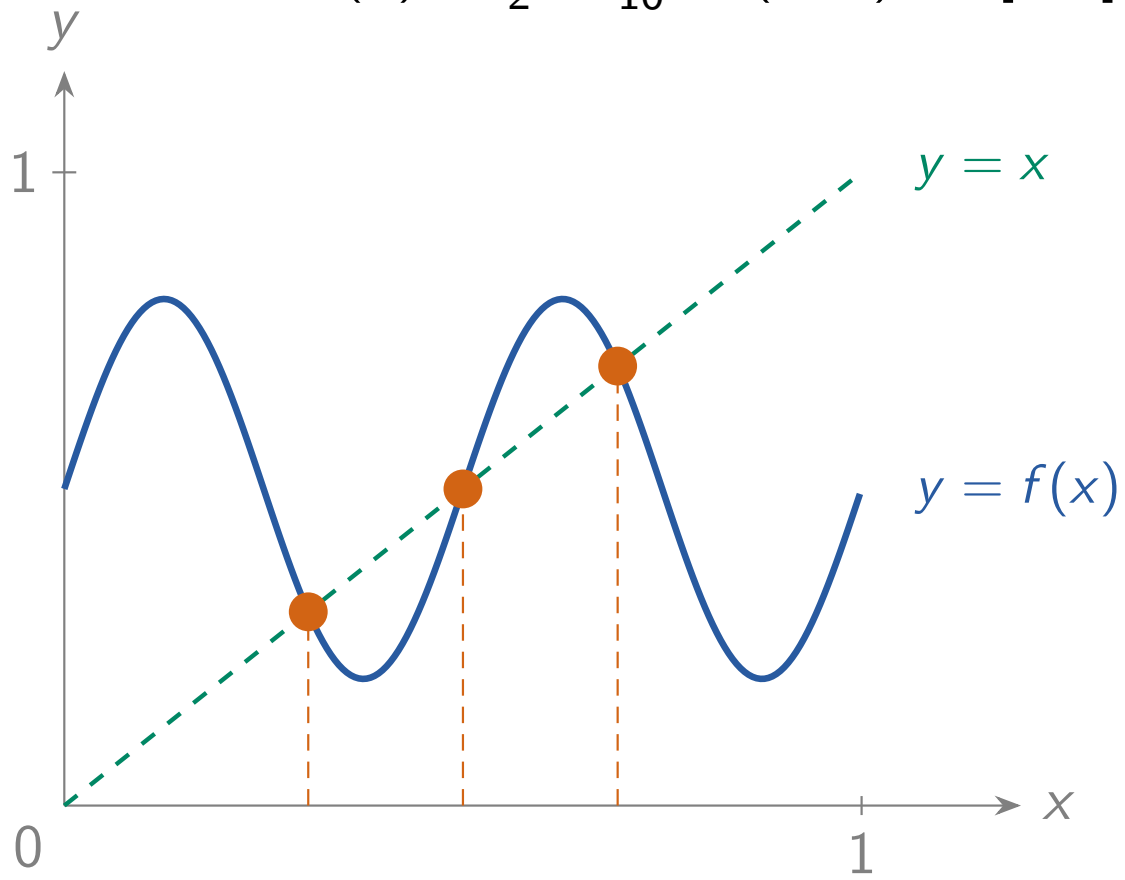
Example: $f(x) = \frac{1}{2} + \frac{3}{10}\sin(4\pi x)$ on $[0, 1]$.



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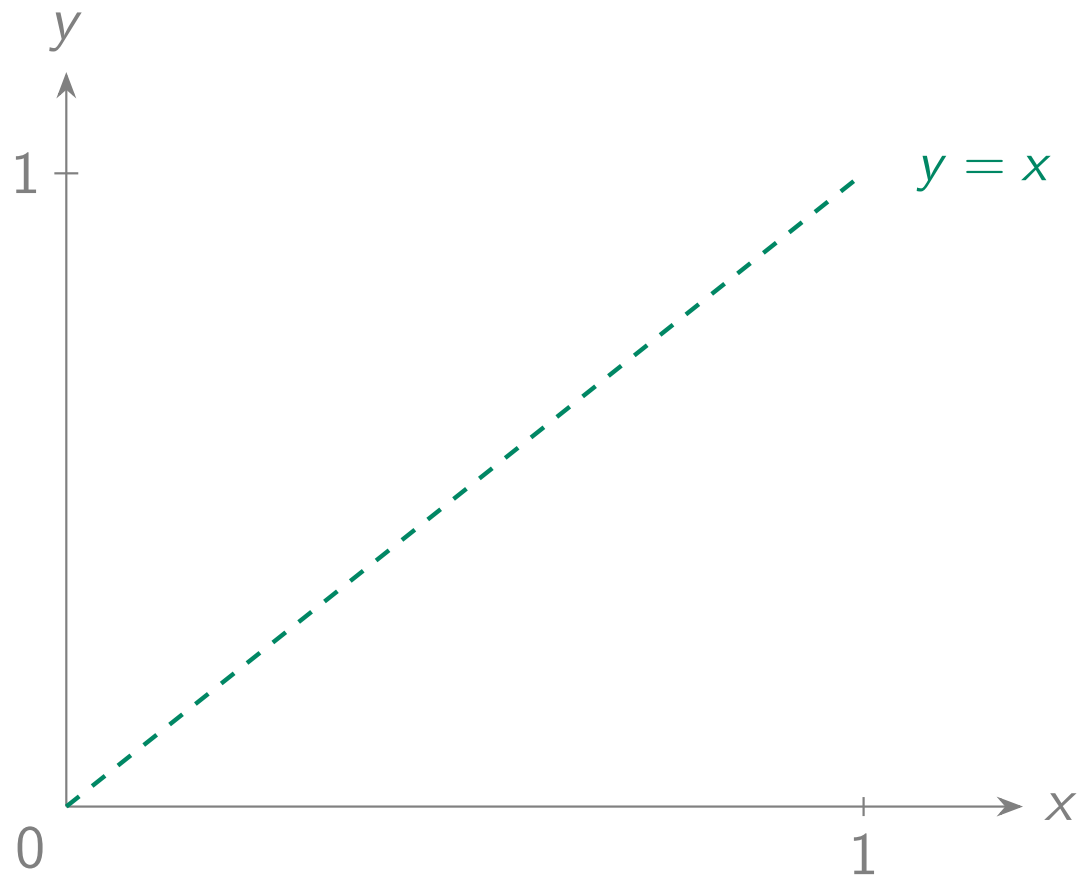
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There can be more than one fixed point.

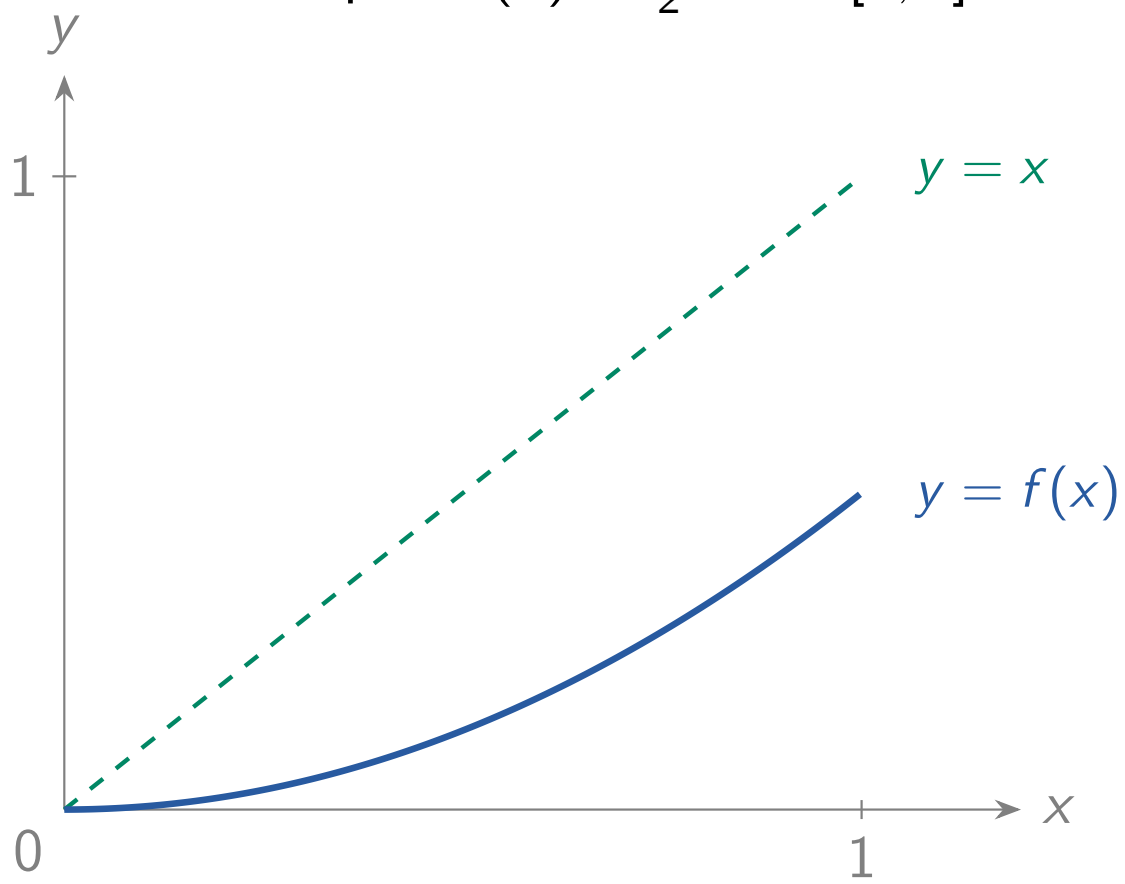
Exercise 6.2: Intuition — Starting at Zero



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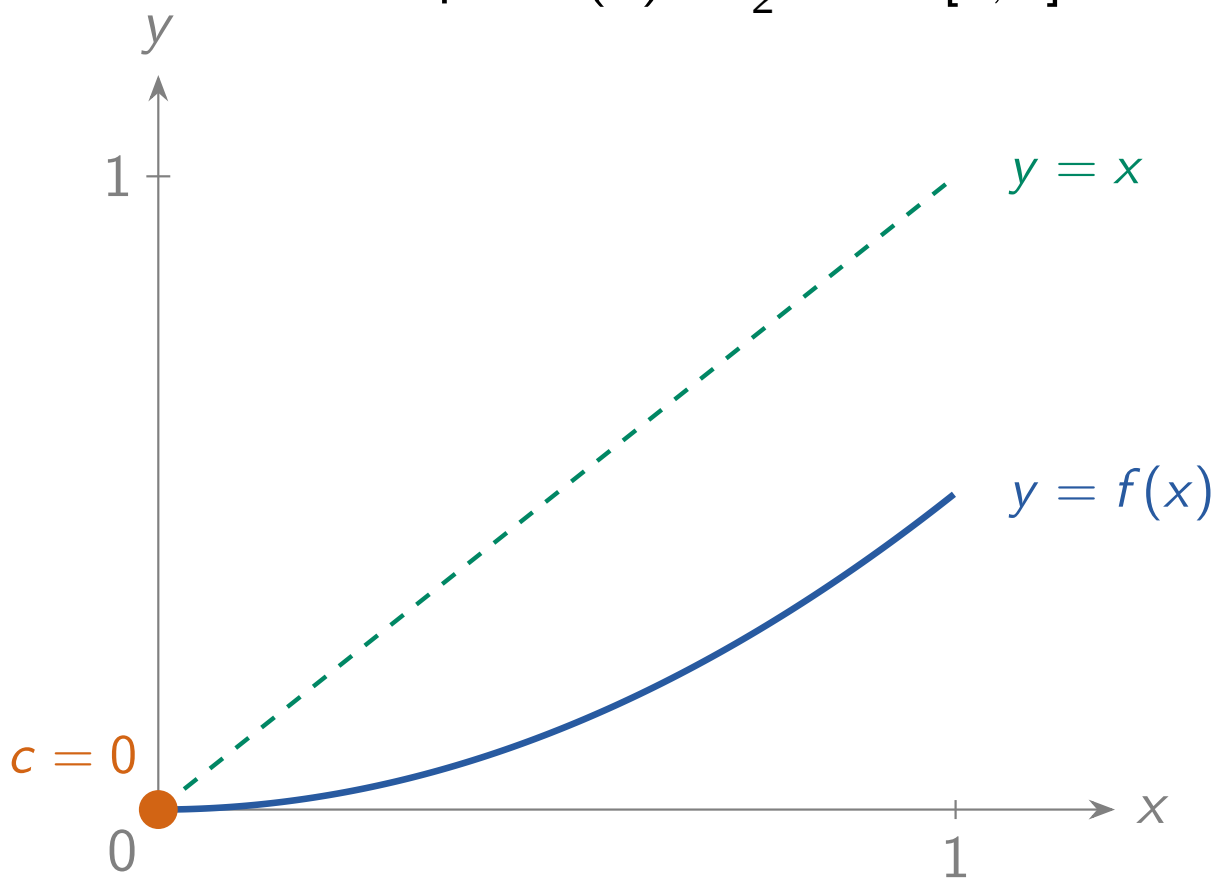
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Where does the graph meet the diagonal?

Exercise 6.2: Intuition — Starting at Zero

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The endpoint itself can be a fixed point: $f(0) = 0$.

Appendix: Open Sets in $[0, 1]$

$$\boxed{U = [0, \tfrac{1}{2}) \text{ is open in } Y = [0, 1].}$$

Open in a subspace

A set $U \subseteq Y$ is open in Y if there exists an open set $O \subseteq \mathbb{R}$ such that $U = O \cap Y$.

1. Here, the codomain is $Y = [0, 1]$.

◂ Exercise 5.5: Lhc

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1. Here, the codomain is $Y = [0, 1]$.
2. Take the open interval $O = (-\frac{1}{2}, \frac{1}{2})$ in $\mathbb{R}$. Then

$$O \cap Y = (-\tfrac{1}{2}, \tfrac{1}{2}) \cap [0, 1] = [0, \tfrac{1}{2}).$$

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$$O \cap Y = (-\tfrac{1}{2}, \tfrac{1}{2}) \cap [0, 1] = [0, \tfrac{1}{2}).$$

3. Thus U is open in Y. Neighborhoods are taken inside $[0, 1]$, including at zero.

◂ Exercise 5.5: Lhc

Appendix: Why the Sequential Condition Implies Uhc

$$\boxed{\text{The sequential condition at } q \implies B \text{ is uhc at } q}$$

1. Suppose uhc fails at q. There is an open $U \subseteq \mathbb{R}^2_+$ containing $B(q)$ such that no neighborhood of q works.

◂ Exercise 5.6: Definitions and tools

Appendix: Why the Sequential Condition Implies Uhc

The sequential condition at $q \implies B$ is uhc at q

1. Suppose uhc fails at q. There is an open $U \subseteq \mathbb{R}^2_+$ containing $B(q)$ such that no neighborhood of q works.
2. For each n, choose $q^n \in \mathbb{R}^3_{++}$ and $y^n \in B(q^n)$ with

$$\rho(q^n, q) < 1/n, \qquad y^n \notin U.$$

Thus $q^n \to q$.

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   Thus $q^n \to q$.
3. By the sequential condition, some subsequence $y^{n_k} \to y \in B(q) \subseteq U$.
4. Since U is open, $y^{n_k} \in U$ for all sufficiently large k. This contradicts the choice of every y^n outside U.

◂ Exercise 5.6: Definitions and tools

Appendix: Why $g(x) = f(x) - x$ Is Continuous

$$\boxed{f \text{ continuous on } [0,1] \implies g \text{ continuous on } [0,1]}$$

1. Fix any $x_0 \in [0,1]$ and any $\epsilon > 0$.

◂ Exercise 6.2: Sketch of the proof

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2. By continuity of f, choose $\delta_f > 0$ such that, for every $x \in [0,1]$,

$$|x - x_0| < \delta_f \implies |f(x) - f(x_0)| < \epsilon/2.$$

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3. Set $\delta = \min\{\delta_f, \epsilon/2\}$. For $x \in [0,1]$ with $|x - x_0| < \delta$,

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3. Set $\delta = \min\{\delta_f, \epsilon/2\}$. For $x \in [0,1]$ with $|x - x_0| < \delta$,

$$\begin{aligned} |g(x) - g(x_0)| &= |(f(x) - f(x_0)) - (x - x_0)| \\ &\leq |f(x) - f(x_0)| + |x - x_0| < \epsilon. \end{aligned}$$

4. Conclude continuity at every $x_0 \in [0,1]$.

◂ Exercise 6.2: Sketch of the proof